Xutao Mao

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Website | GitHub | LinkedIn

Education

- **City University of Hong Kong** — *Hong Kong*
  Ph.D. Student, Computer Science — Sep 2026 – Aug 2030 (Expected)
  - Research Areas: AI Safety

- **Vanderbilt University** — *Nashville, US*
  Bachelor of Science, Honors in Computer Science & Mathematics — Aug 2022 – Dec 2025
  - Research Areas: LLM, Data Mining

Research Experience

- **City University of Hong Kong** — *Hong Kong*
  Graduate Researcher | Advisor: Dr. Cong Wang — Aug 2026 – Present
  - Acting and coordinating: how AI fails while they act in the world [AHA] [CAVE-Bench] [TrustFork]
  - Remembering and improving: how AI failures persist [PASB] [MisEvolve]
  - Looking inside and stepping in: how AI failures happens & how to mitigate failures [STARE] [Memory Interpret]

- **Hong Kong Baptist University** — *Hong Kong*
  Research Assistant | Advisor: Dr. Bo Han — Dec 2025 – Aug 2026
  - Focus: CoT Monitorability [TAME]

- **Vanderbilt University** — *Nashville, US*
  Undergraduate Researcher | Advisor: Dr. Tyler Derr — Feb 2025 – Dec 2025
  - Focus: LLM; Data Mining [Review Recommendation]

Publications

C=Conference, P=Preprint

[C.1] **Xutao Mao**[*], Jianing Zhu[*], Jinman Zhao, Tongliang Liu, Xiaowen Chu, Cong Wang[‡], Bo Han[‡]. **Taming CoT Obfuscation in VLMs: From Mechanistic Evidence to Activation-Level Enforcement.** *NeurIPS 2026* **(Poster)**.

[C.2] **Xutao Mao**, Liangjie Zhao, Tao Liu, Xiang Zheng[‡], Hongying Zan, Cong Wang[‡]. **STARE: Step-wise Temporal Alignment and Red-teaming Engine for Multi-modal Toxicity Attack**. *ICML 2026* **(Poster)**.

[C.3] Haobo Zhang[*], **Xutao Mao**[*], Guangyuan Dong, Ziwei Li[‡], Xuanbo Su, Kaijie Chen, Jing Yang, Zheng Lin. **MemMark: State-Evolution Attribution Watermarking for Agent Long-Term Memory Systems**. *EMNLP 2026* **(Finding)**.

[C.4] **Xutao Mao**[‡], Ezra Tao, Leyao Wang. **MindVote: When AI Meets the Wild West of Social Media Opinion**. *AAAI 2026* **(Oral)**.

[C.5] Tao Liu[†], **Xutao Mao**[†], Hongying Zan[‡], Dixuan Zhang, Yifan Li, Haixin Liu, Lulu Kong, Jiaming Hou, Rui Li, YunLong Li, Aoze Zheng, Zhiqiang Zhang, Zhewei Luo, Kunli Zhang, Min Peng. **LogicCat: Text-to-SQL Benchmark for Multi-Domain Reasoning Challenges**. *AAAI 2026* **(Poster)**.

[P.1] **Xutao Mao**, Rui Qian, Xiang Zheng[‡], Cong Wang[‡]. **Agent Hacks Agents: Autoresearch Discovers Vulnerabilities in Production Agents**.

[P.2] **Xutao Mao**[*], Rui Qian[*], Longxiang Wang, Xinjian Yi, Mingxuan Li, Linghan Chen, Yudong Gao, Xiang Zheng[‡], Cong Wang[‡]. ***You're Right, Let Me Fix It*: How LLM Agents Damage Correct Work When Falsely Accused**.

[P.3] **Xutao Mao**[*], Rui Qian[*], Linghan Chen, Yudong Gao, Junchi Liao, Jiulin Cai, Jinman Zhao, Cong Wang[‡]. **Trust the Brand, Lose Control: How Identity Hijacks LLM Agent Orchestration**.

[P.4] **Xutao Mao**[*], Liangjie Zhao[*], Leyao Wang, Rui Qian, Qiang Huang, Wentao Wang, Bo Han[‡], Xiang Zheng[‡], Cong Wang[‡]. **Agents Don't Just Agree, They Remember: Benchmarking Persistent Sycophancy in Self-Improving Personal Agents**.

[P.5] **Xutao Mao**, Liangjie Zhao, Xiang Zheng[‡], Cong Wang[‡]. **Practice Makes Unsafe: Skill Misevolution in Self-Improving LLM Agents**.

[P.6] **Xutao Mao**, Jinman Zhao, Gerald Penn, Cong Wang[‡]. **What Happens Inside Agent Memory? Circuit Analysis from Emergence to Diagnosis**.

[P.7] Leyao Wang[*], **Xutao Mao**[*], Xuhui Zhan, Yuying Zhao, Bo Ni, Ryan A. Rossi, Nesreen K. Ahmed, Tyler Derr[‡]. **Towards Bridging Review Sparsity in Recommendation with Textual Edge Graph Representation**.

[P.8] Wenjie Lin[*], Hange Liu[*], Yingying Zhuang, **Xutao Mao**, Jingwei Shi, Xudong Han, Tianyu Shi, Jinrui Yang[‡]. **ParlAI Vote: A Web Platform for Analyzing Gender and Political Bias in Large Language Models**.

[P.9] Yudong Gao*, Linghan Chen*, Wenhan Wu, Quan Shi, **Xutao Mao**, Mia Zhou, Junjian Li, Xiaolong Liu, Jiyao Wang, Mingyu Guo, Honglong Chen‡. **Cooldown Landmines: Cross-Tenant Interference Attacks on LLM Gateways**.

Notes: *equal contribution; † alphabetical co-first/co-equal as noted; ‡corresponding author.

Work Experience

- **Pegasystems** *Cambridge, US*
  Software Development Engineer Intern May 2024 – Aug 2024

Honors and Awards

- **Hong Kong Postgraduate Scholarship** *2026*
  City University of Hong Kong

- **Vanderbilt Summer Research Program Scholarship** *2025*
  Vanderbilt University

Professional Service

- **Conference Reviewer:** ICLR '27; NeurIPS '26; AAAI '26 / '27; TheWebConf '26; ACL ARR (Oct 26)

Technical Skills & Interests

- **Models & Training:** LLM/VLM architectures (MoE, RoPE, FlashAttention); SFT, RL post-training (PPO, GRPO, DPO, RLHF), agentic RL, LoRA; PyTorch, DeepSpeed, FSDP, vLLM, verl; multi-node multi-GPU training
- **Research Interests:** AI Safety, Interpretability.